Where Does the Estimation Error Go?

Markowitz, RMT-denoised, PCA-factor and HRP portfolios on daily S&P 500 data

DSE4212/QF4212 Project 2 — Group [number and names to be filled in]

15 November 2026

1 Overview of the problem

Mean-variance portfolios built from sample moments perform poorly out of sample: the optimizer puts its largest positions where the estimation error is largest [10, 8]. The project asks *where the estimation error goes*, and whether methods that clean the covariance matrix give better out-of-sample portfolios than the sample-covariance plug-in.

In one rolling backtest with identical universe, dates, constraints and costs, we compare the sample covariance, Ledoit–Wolf shrinkage [4], RMT eigenvalue clipping [9, 3], a PCA factor model and hierarchical risk parity (HRP, [6]), with 1/N as the naive benchmark [1]. Two hypotheses were fixed beforehand: **H1**, denoised covariances give lower out-of-sample volatility than the sample covariance in the global minimum-variance (GMV) portfolio; **H3**, the advantage of denoising is larger without the long-only constraint, because the constraint already regularizes [2]. Parameters were fixed a priori and never tuned on the test period; null results are reported as results. All figures and tables are collected in the Appendix.

Summary of findings. Equal weight has the best Sharpe ratio, and no Sharpe difference is statistically detectable. GMV cuts volatility by about a quarter relative to 1/N, but denoising does *not* improve on sample GMV (PCA and RMT are worse), so H1 is not supported; H3 holds only for Ledoit–Wolf. In a simulation with a known factor covariance the denoisers do win, by only about 1% of variance, and the advantage disappears when the covariance is time-varying. The extensions (Section 5.7) do not change the picture.

2 Dataset description

We use only the supplied Kaggle files of daily S&P 500 prices. The stock file lists 502 symbols, but only 172 have prices (the other 330 are empty; Apple, for example, has none). Cleaning is done in code: symbols without prices are dropped and logged, and 11 suspect returns across 9 symbols are flagged. The cleaned panel runs from 2010-01-05 to 2024-12-20 (3,767 return dates). The index file is used only for a comparison on its common period, and sector labels only as labels.

At each rebalance date the universe is the top $N = 100$ names by trailing 504-day dollar volume among symbols with a complete $T = 504$-day history, using data up to that date only (150 to 170 symbols are eligible). The first window ends 2012-01-03, so the out-of-sample period is 2012-01-04 to 2024-12-20 (3,263 trading days, 156 monthly rebalances). The 172 priced names are current index members, so the universe carries survivorship bias; relative comparisons share it. $N = 200$, a planned sensitivity value, is infeasible.

3 Initial data analysis

Daily returns are strongly fat-tailed: on the first, stress and last snapshot windows the pooled excess kurtosis is 11.7, 25.9 and 29.2 (Figure 1), so windows are winsorized per column (0.1%/99.9%) before estimation.

For N assets and T observations of pure noise, the eigenvalues of a correlation matrix lie below the Marchenko–Pastur (MP) edge $\lambda_+ = (1 + \sqrt{q})^2$, $q = N/T$ [9]; here $q = 0.20$ and $\lambda_+ = 2.09$. Figure 2 shows that only 2, 4 and 6 eigenvalues rise above the edge at the three snapshots, the largest (the market mode) taking 53%, 49% and 23% of the total; the rest is a bulk that the sample estimate spreads too widely, which is what RMT clipping targets. Across the 156 windows, the number of signal eigenvalues ranges from 2 to 6 (median 4) and the average correlation from 0.21 to 0.52, highest at the start (Figure 3). The 21-day average correlation is 0.32 over the sample, 0.38 in 2020 and 0.43 in 2022: correlations jump in stress periods, so an estimator calibrated on a calm window is exposed.

4 Methods used

The central tools are unsupervised learning methods (eigen-decomposition and PCA, random matrix theory, hierarchical clustering) that build the covariance input of the optimizer. Supervised learning (Ridge and random forest) enters in one extension.

Backtest. Log returns are used for estimation and simple returns for profit and loss. Weights at rebalance date t use only data up to t (including the universe and winsorization) and apply from $t + 1$, drift between rebalances and pay 10 bps on turnover. Every strategy is long-only, fully invested and faces the same universe, dates and costs. Unit tests check the absence of look-ahead and the accounting.

Covariance estimators (written by hand in NumPy/SciPy). *Sample* covariance S. *Ledoit–Wolf*: $\hat{\mathbf{\Sigma}} = (1-\delta)S + \delta(\operatorname{tr} S/N)I$ with a data-driven δ [4]. *RMT clipping*: correlation eigenvalues above λ_+ are kept, the rest replaced by their mean (trace preserved), rescaled to a unit diagonal [3]. *PCA factor model*: $\hat{\mathbf{\Sigma}} = B\Lambda_K B^\top + \text{diag(residual)}$ with K from the MP rule. *HRP*: clustering of the correlation matrix and recursive bisection by cluster variance, with no matrix inversion [6].

Portfolios. 1/N; GMV, $\min_w w^\top \hat{\boldsymbol{\Sigma}} w$ s.t. $\mathbf{1}^\top w = 1$, $w \geq 0$, for each covariance; HRP; and long-only maximum-Sharpe (Markowitz) with sample and RMT covariance, using window means (mean log return plus half the variance) as expected returns and a zero risk-free rate.

Evaluation. Annualized return, volatility, Sharpe, Sortino, drawdown, CVaR, turnover and the effective number of names $1/\sum_i w_i^2$, net and gross of costs, plus the volatility the optimizer *predicts* against the one realized over the next 21 days. Differences are tested with a studentized circular block bootstrap (block 10, 5,000 resamples, strategies resampled jointly; Sharpe differences and log variance ratios), Holm adjustment within each family and the smallest effect detectable at 80% power.

Simulation and extensions. A factor-model simulation with known covariance ($N = 100$, three true factors, 200 replications; stationary and time-varying) separates "the estimator is worse" from "the data are different". Extensions use the same engine: inverse-volatility and equal-risk-contribution (ERC) portfolios [7]; ridge, shrinkage-to-1/N, L1-bounded and elastic-net weights; quadratic-inverse nonlinear shrinkage [5]; EWMA covariances; a sector-block residual covariance with a random-label placebo; and the QLIKE forecast loss of each covariance on random portfolios (Newey–West errors, [11]).

Machine-learning return forecasts. Ridge and a shallow random forest (200 trees, depth 4) predict the cross-sectional rank of the next 21-day return from 12-1 month momentum, 1-month return, 63-day volatility and dollar volume. Training uses an expanding window, a refit every six rebalances and a purge so no training label reaches past the refit date. The signal is judged by its out-of-sample rank information coefficient (IC); a pre-set gate (mean IC > 0 and $t \geq 2$) decides whether it may feed the optimizer, as a centered rank score with the RMT covariance. All hyperparameters were fixed before the run.

5 Model performance evaluation and application results

5.1 Estimated versus realized Markowitz frontier

Plug-in Markowitz treats estimates as truth. At the first window (2012-01-03) the maximum-Sharpe portfolio has an estimated Sharpe ratio of 2.07 with the sample covariance against 0.87 realized over the next 252 days; at the latest date with a full year ahead (2023-12-20) the figures are 1.87 and 1.01 (Figure 4). The stress date (2020-03-31) is an exception because of the rebound that followed. The RMT covariance does not change the picture, which points to expected returns as the dominant error in the Markowitz problem.

5.2 Out-of-sample performance

Table 1 and Figure 5 give the main result. Equal weight has the highest Sharpe ratio (0.99) and HRP is next (0.93 to 0.94). GMV portfolios have the lowest volatility (about 0.13 against

0.17 for 1/N) but also the lowest return, so their Sharpe ratios are lower (0.66 to 0.76). The Markowitz portfolios earn a return close to 1/N with higher volatility (0.20) and a turnover of about 6 a year, against 0.9 for 1/N and 2.2 to 2.7 for GMV and HRP. Costs of 10 bps lower Sharpe ratios by only 0.01 to 0.03 (Table 2). Denoising does not lower GMV volatility: RMT (0.136) and PCA (0.135) are above sample GMV (0.130), and Ledoit–Wolf matches it. GMV portfolios are concentrated (about 10 effective names against about 66 for HRP, Table 3) and tilt towards Consumer Defensive and Utilities and away from Technology (Figure 13).

5.3 Statistical evidence

All GMV variants reduce volatility relative to 1/N by 22 to 25% (significant after Holm). Against sample GMV, PCA has 3.3% higher volatility (95% interval 2.4 to 4.3%, significant), Ledoit–Wolf is 0.6% lower (significant but negligible), and RMT is 4.5% higher with an interval that includes zero in the studentized test ($p = 0.060$). The RMT result is therefore suggestive, not established: its Holm-adjusted p-value is 0.022, 0.060 and 0.201 for block lengths 5, 10 and 20, while all other conclusions are stable. The Markowitz portfolios have about 1.5 times the volatility of sample GMV.

No Sharpe difference is significant, but this is mostly a power problem, not evidence of equality: the smallest detectable difference is about 0.5 to 0.6 against 1/N and 0.13 to 0.17 against sample GMV.

5.4 Does the optimizer know its own risk?

The pooled ratio of realized to predicted variance is 1.32 for sample GMV, 1.39 for RMT and 1.61 for PCA, so the optimizers under-predict their own risk, and the denoised versions by more; HRP does not (0.97) (Figure 6, Table 4). This is consistent with an optimizer that chooses weights against its own estimate, an effect a better-conditioned covariance does not remove.

5.5 Subperiods, sensitivity and the long-only constraint

Rankings are not stable across subperiods (Figure 7). Minimum-variance portfolios have a Sharpe ratio near zero in the COVID stress period (0.06 to 0.08 against 0.67 for 1/N) while the Markowitz portfolios exceed 1.5; in 2022 every strategy is negative, GMV-PCA the least (-0.02); in calm markets 1/N and HRP lead. A 250-day subperiod has a Sharpe standard error near 1, so this is descriptive. Across 15 sensitivity scenarios (81 strategy-parameter combinations, reported as sensitivity and never used to choose a configuration), one result is stable: at every $q = N/T$ from 0.10 to 0.40, GMV-RMT and GMV-PCA have higher volatility than sample GMV (Figure 8). The loss does not go away with more data per asset.

H3 (Table 5) holds only for Ledoit–Wolf (difference -0.055, interval -0.080 to -0.033). For RMT there is no difference, and PCA goes the other way: it is worse than sample GMV, more so unconstrained. Without the constraint, leverage is 2.7 to 3.7, turnover 6 to 12 times higher and volatility not lower for any estimator (shorts carry no borrow cost here, which flatters them). The long-only constraint is already a strong regularizer [2], leaving little for denoising to add.

5.6 What a known-truth simulation says

On real data the true covariance is unknown, so Figure 11 repeats the comparison with a known factor covariance. With a stationary covariance, PCA and RMT beat sample GMV, by the most at small T (log true-variance ratio about -0.06 at $T = 126$) but only -0.012 (PCA) and -0.005 (RMT), about 1% of variance, at $T = 504$; without the long-only constraint the gain is far larger (about -1.3 at $T = 126$). With time-varying volatilities and loadings the advantage vanishes for $T \geq 504$: the error is mostly non-stationarity bias, which denoising cannot remove. The simulated loss is only about 0.7% of variance, however, far from the 3 to 5% of volatility (about 7 to 9% of variance) seen on real data. The engine is not the cause: run unchanged on synthetic stationary panels, it agrees with the direct evaluation to 10^{-15} and reproduces the simulation's ranking and magnitudes.

5.7 Why does denoising not help? Extensions

Non-stationarity. EWMA covariances (half-lives 126 and 63 days, fixed in advance) win in the simulated time-varying world; on real data their point estimates beat sample GMV (volatility ratio 0.96, Sharpe 0.83 against 0.74) but not significantly, as the test could only detect a 5 to 10% change. Non-stationarity is consistent with the evidence but not confirmed, and it does not explain the RMT/PCA loss (EWMA-RMT is again no better than EWMA).

Sector structure in the residual. The PCA residual still holds sector correlation: the same-sector minus cross-sector gap is 0.061 against 0.000 for shuffled labels (permutation $p < 0.05$ in all 156 windows, descriptive), so the factors remove only about 45% of the sector gap. Restoring same-sector blocks lowers PCA's volatility by 2.4% (Holm $p = 0.001$), removing about three quarters of its loss, but stays above sample GMV (ratio 1.009, Holm $p = 0.036$). True sectors beat all 20 random-label draws (rank $p = 0.048$, the smallest possible), and random labels recover only 18% of the gain. Sector structure is a real but incomplete part of the PCA loss.

Forecast quality versus use. As forecasters of the variance of random long-only portfolios, RMT and PCA are not worse than the sample covariance: QLIKE differences are not significant after Holm (smallest adjusted $p = 0.77$), and the point estimates rank PCA-sector, PCA, RMT, sample, nonlinear shrinkage, Ledoit–Wolf identically in three portfolio families (Table 11). This suggests that the GMV loss comes from how the optimizer exploits the estimate, not from a worse covariance as such. The mechanism was not tested directly, and the test has little power.

Other optimizers and regularization. ERC portfolios are almost insensitive to the covariance estimator (differences below 1%) and have realized-to-predicted volatility of about 1.0 against 1.19 to 1.32 for GMV, so the denoising loss is specific to the minimum-variance optimizer, which exploits the small eigenvalues (Table 13). Ridge and shrinkage towards 1/N push realized volatility towards the predicted level (1.19 to 1.01) at the price of higher volatility, with no significant Sharpe gain (Table 14). Nonlinear shrinkage, the theoretical counterpart of RMT clipping, does not beat sample GMV either (volatility ratio 0.996, Holm $p = 0.12$).

Machine-learning return forecasts. Both models fail the pre-set gate (Table 15): the mean rank IC is 0.025 for Ridge ($t = 1.25$) and 0.029 for the random forest ($t = 1.70$) over 131 dates, against a gate of $t \geq 2$, and the single features are no better. The portfolios built on the forecasts are shown for completeness (Table 16): the optimizer turns a noise ranking into a concentrated, high-turnover portfolio (about 5 effective names, turnover 17 to 19 a year) with volatility 15% to 22% above MV-RMT (Holm $p = 0.02$ and 0.001) and a lower Sharpe ratio. This is a null result. The centered-rank mapping drops the market premium that MV-RMT keeps, so part of the gap may come from the mapping, which was fixed before the results.

6 Conclusion

Answer to the research question. On this data the estimation error does *not* go where denoising can remove it. Cleaning the covariance matrix reduces condition numbers by a factor of three to six, yet does not lower the out-of-sample volatility of GMV and, for PCA, raises it. Our evidence points to three parts of the explanation, none isolated fully: sector correlation that denoising throws away (real but incomplete), non-stationarity of the covariance (consistent with the simulation, underpowered on real data), and the optimizer exploiting its estimate (covariance forecasts are not worse, but GMV portfolios under-predict their own risk). The long-only constraint already regularizes much of the problem. In risk-adjusted terms 1/N is hard to beat: GMV gives the lowest volatility at a lower Sharpe ratio, while ERC and HRP give diversified portfolios with a Sharpe ratio close to 1/N.

Limitations. (i) Only 172 of 502 symbols have prices and they are current members, so survivorship bias remains, the sample is mostly a bull market, and $N = 200$ was infeasible. (ii) One historical path: the bootstrap quantifies sampling noise within it, and many differences are inside the detectable-effect bounds, so "not significant" means "not detectable". (iii) Costs are proportional, with no market impact and no borrow cost for shorting variants. (iv) The ML evaluation starts two years after the core backtest (132 dates), uses four features and treats the 21-day-spaced IC values as independent. (v) The "optimizer selects against the estimate" mechanism rests on indirect evidence, and the sector and non-stationarity diagnostics are descriptive with overlapping windows.

Future work. Test the optimizer mechanism directly (for example by resampling diagnostics); repeat the study on a full point-in-time S&P 500 file with delistings; combine time-varying volatility with denoising; denoise eigenvectors as well as eigenvalues; and try richer return signals with a longer training history.

Reproducibility. Parameters are in `config.yaml`; the submitted code regenerates every table and figure.

References

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A Appendix: figures and tables

All tables are also available as CSV, Markdown and LaTeX in `results/report/tables`, and all figures as PDF and PNG in `results/report/figures`. The long tables (bootstrap tests, subperiods, sensitivity, the parameter table) are not repeated here; see `tab03_significance`, `tab04_subperiods`, `tab05_sensitivity`, `tab08_parameters` and `tabA09_inference`.

A.1 Data analysis (Section 3)

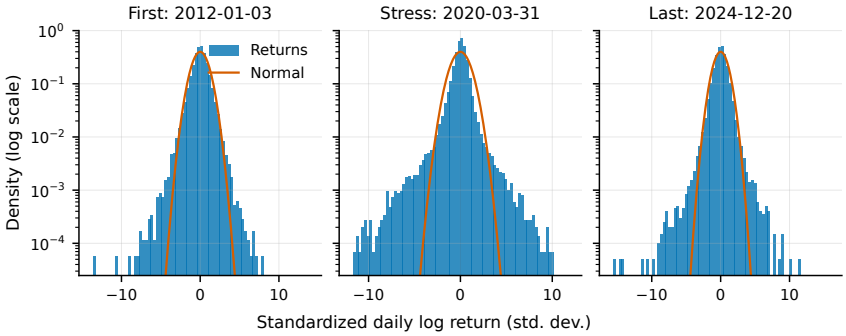


Figure 1: Pooled return distribution.

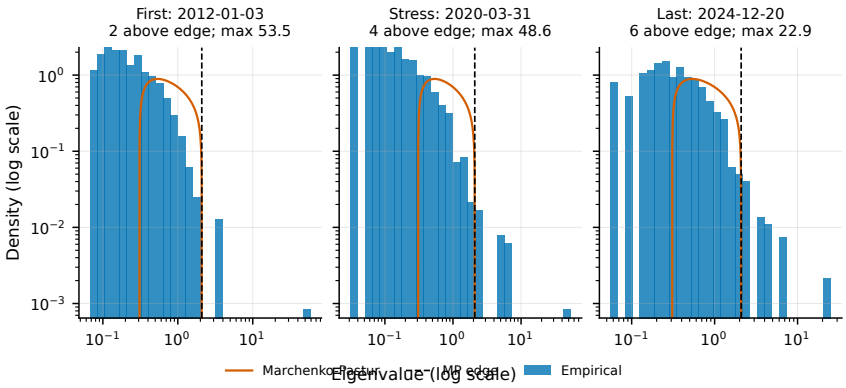


Figure 2: Eigenvalue spectrum of the sample correlation matrix against the Marchenko–Pastur density (first, stress and last snapshot windows).

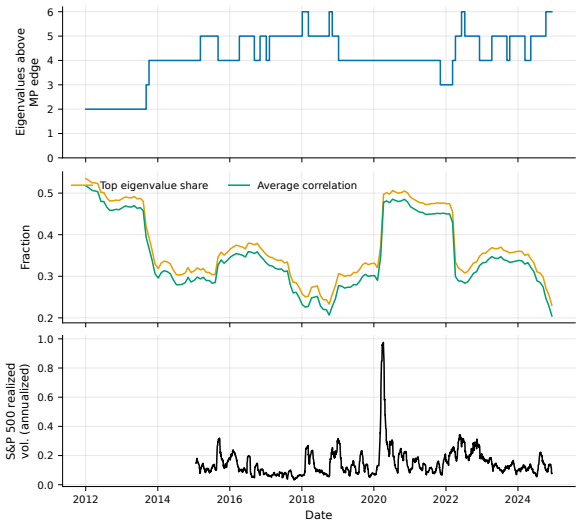


Figure 3: Number of eigenvalues above the MP edge and the market-mode share over time.

A.2 Main results (Sections 5.1 to 5.6)

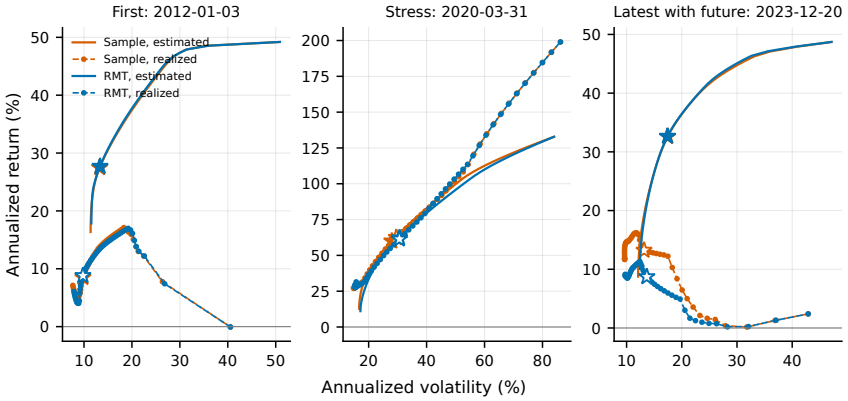


Figure 4: Estimated against realized efficient frontier at three dates (sample and RMT covariance).

Table 1: Out-of-sample metrics, net of costs.

strategy	ann_return	ann_vol	sharpe	sortino	max_drawdown	calmar	cvar_95	annual_turnover
EW	0.170	0.174	0.993	1.401	-0.376	0.453	0.026	0.869
GMV-LW	0.094	0.130	0.758	1.066	-0.335	0.280	0.019	2.262
GMV-PCA	0.092	0.135	0.721	1.013	-0.343	0.268	0.020	2.175
GMV-RMT	0.084	0.136	0.663	0.932	-0.362	0.233	0.020	2.195
GMV-SAMPLE	0.093	0.130	0.744	1.047	-0.333	0.278	0.019	2.397
HRP-RMT	0.138	0.150	0.941	1.316	-0.359	0.385	0.022	2.535
HRP	0.136	0.148	0.934	1.307	-0.356	0.382	0.022	2.737
MV-RMT	0.174	0.202	0.898	1.291	-0.383	0.455	0.030	6.029
MV-SAMPLE	0.177	0.199	0.919	1.315	-0.380	0.466	0.030	5.812

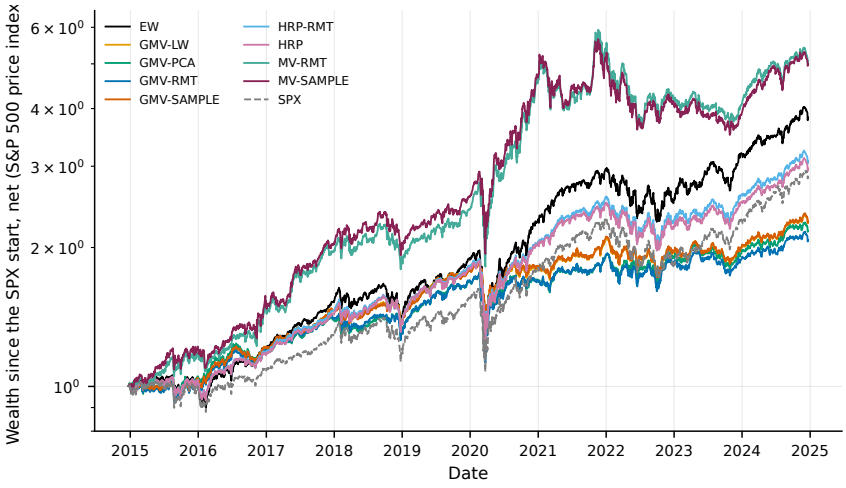


Figure 5: Net equity curves of the core strategies, 1/N and the S&P 500 (common period).

Table 2: Out-of-sample metrics, gross of costs.

strategy	ann_return	ann_vol	sharpe	sortino	max_drawdown	calmar	cvar_95	annual_turnover
EW	0.171	0.174	0.998	1.409	-0.376	0.456	0.026	0.869
GMV-LW	0.097	0.130	0.776	1.092	-0.335	0.288	0.019	2.262
GMV-PCA	0.094	0.135	0.738	1.037	-0.343	0.275	0.020	2.175
GMV-RMT	0.087	0.136	0.680	0.956	-0.362	0.240	0.020	2.195
GMV-SAMPLE	0.095	0.130	0.763	1.075	-0.333	0.286	0.019	2.397
HRP-RMT	0.141	0.150	0.958	1.342	-0.359	0.393	0.022	2.535
HRP	0.139	0.149	0.953	1.335	-0.356	0.391	0.022	2.737
MV-RMT	0.181	0.202	0.928	1.336	-0.376	0.483	0.030	6.029
MV-SAMPLE	0.184	0.199	0.948	1.359	-0.372	0.494	0.030	5.812

Table 3: Weight concentration.

strategy	mean_hhi	mean_effective_n	mean_max_weight	mean_top10_share	mean_n_positions	share_zero	p50	p90	p99	max
EW	0.010	100.000	0.010	0.100	100.000	0.000	0.010	0.010	0.010	0.010
GMV-SAMPLE	0.112	10.275	0.207	0.844	20.109	0.799	2.3e-12	0.030	0.158	0.430
GMV-LW	0.094	11.888	0.177	0.809	21.994	0.780	2.6e-12	0.035	0.142	0.354
GMV-RMT	0.112	10.008	0.202	0.859	19.763	0.802	2.7e-12	0.029	0.160	0.417
GMV-PCA	0.101	10.690	0.178	0.852	19.538	0.805	2.2e-12	0.032	0.151	0.294
HRP	0.015	66.327	0.039	0.266	99.865	0.001	0.008	0.019	0.037	0.064
HRP-RMT	0.015	68.381	0.038	0.257	99.955	4.5e-04	0.008	0.019	0.036	0.061
MV-SAMPLE	0.157	6.932	0.256	0.967	11.365	0.886	1.5e-13	0.014	0.208	0.512
MV-RMT	0.162	6.841	0.263	0.962	11.359	0.886	1.6e-13	0.014	0.214	0.481

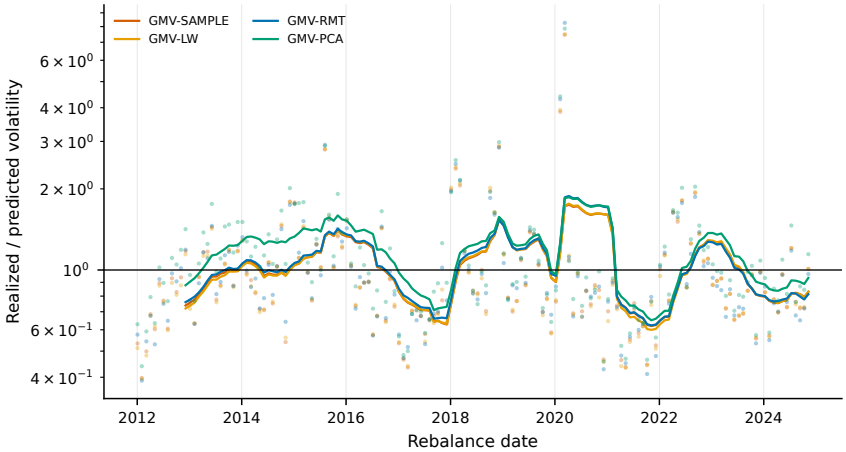


Figure 6: Realized over predicted volatility at each rebalance, by strategy.

Table 4: Volatility forecast quality.

strategy	n_periods	mean_ratio	median_ratio	mean_log_ratio	pooled_variance_ratio	bias_statistic	realized_vol_pooled	mean_predicted_vol
GMV-SAMPLE	155	1.020	0.860	-0.094	1.319	1.227	0.130	0.113
GMV-LW	155	1.018	0.860	-0.099	1.314	1.230	0.130	0.113
GMV-RMT	155	1.039	0.877	-0.079	1.393	1.276	0.136	0.115
GMV-PCA	155	1.147	0.988	0.028	1.605	1.356	0.135	0.106
HRP	155	0.899	0.733	-0.252	0.972	1.105	0.148	0.151
HRP-RMT	155	0.891	0.718	-0.262	0.953	1.094	0.150	0.153
MV-SAMPLE	155	1.029	0.896	-0.067	1.122	1.180	0.199	0.188
MV-RMT	155	1.016	0.880	-0.081	1.065	1.173	0.202	0.195

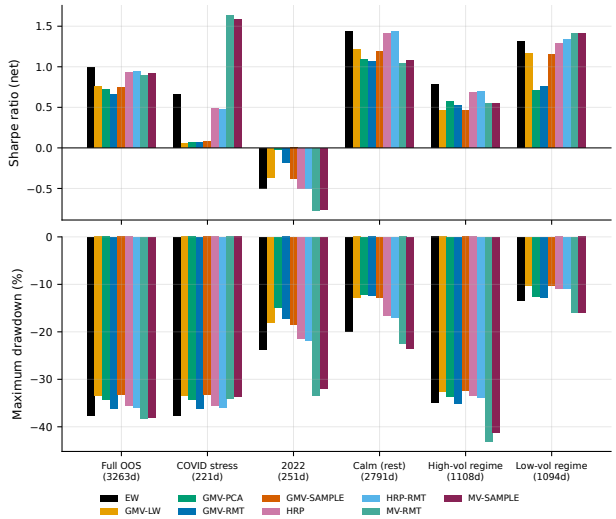


Figure 7: Net Sharpe ratio and maximum drawdown by subperiod.

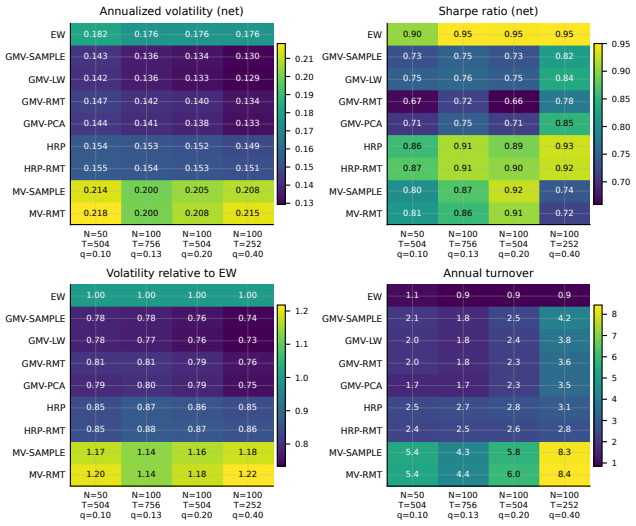


Figure 8: Sensitivity to the ratio $q = N/T$ (common window).

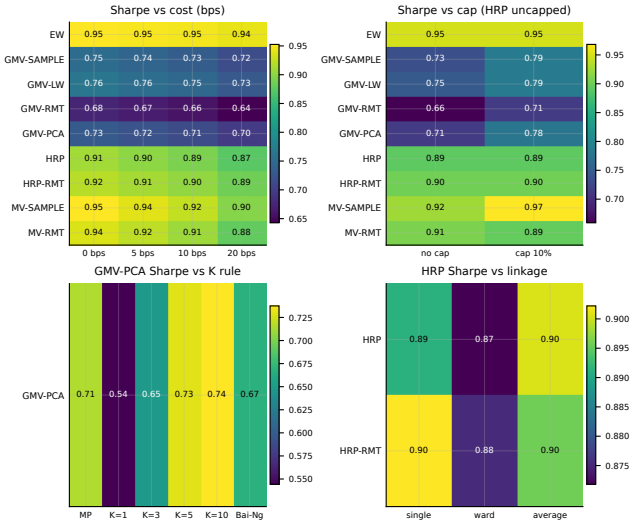


Figure 9: Sensitivity to costs, weight cap, number of PCA factors and HRP linkage.

Table 5: Hypothesis H3 bootstrap test.

name	gap_a	gap_b	estimate	ci_low	ci_high	p_value	significant
Ledoit-Wolf	-0.068	-0.012	-0.055	-0.080	-0.033	4.0e-04	yes
RMT	0.090	0.087	0.003	-0.050	0.049	0.952	no
PCA	0.168	0.066	0.103	0.054	0.161	4.0e-04	yes

Table 6: Long-only and unconstrained GMV.

strategy	constraint	ann_vol	sharpe	annual_turnover	mean_max_weight	mean_leverage	realized_over_pred
GMV-SAMPLE	long-only	0.130	0.744	2.397	0.207	1.000	1.187
GMV-LW	long-only	0.130	0.758	2.262	0.177	1.000	1.185
GMV-RMT	long-only	0.136	0.663	2.195	0.202	1.000	1.226
GMV-PCA	long-only	0.135	0.721	2.175	0.178	1.000	1.319
GMV-SAMPLE-UNC	unconstrained	0.134	0.702	12.061	0.190	3.663	1.551
GMV-LW-UNC	unconstrained	0.130	0.757	9.498	0.149	3.146	1.455
GMV-RMT-UNC	unconstrained	0.140	0.532	6.700	0.161	2.723	1.552
GMV-PCA-UNC	unconstrained	0.146	0.551	6.289	0.128	2.670	1.989

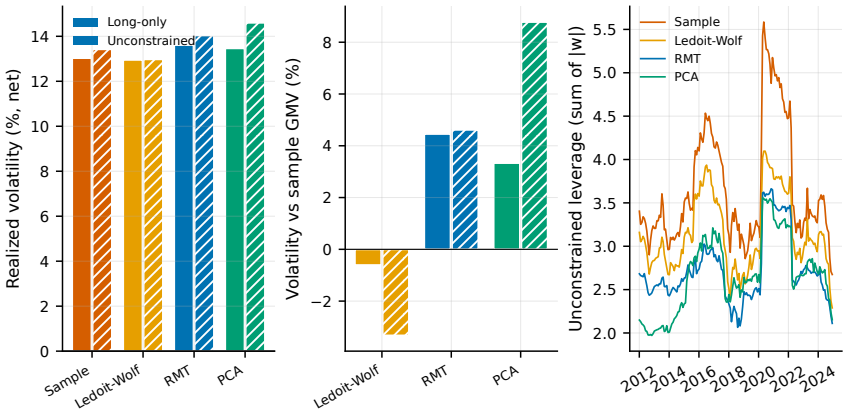


Figure 10: Long-only against unconstrained GMV (hypothesis H3).

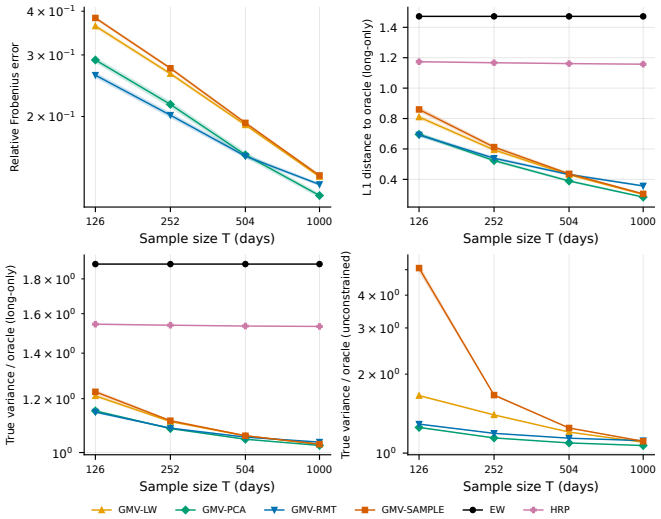


Figure 11: Simulation study with Gaussian returns: true variance of each GMV portfolio relative to the oracle, by window length.

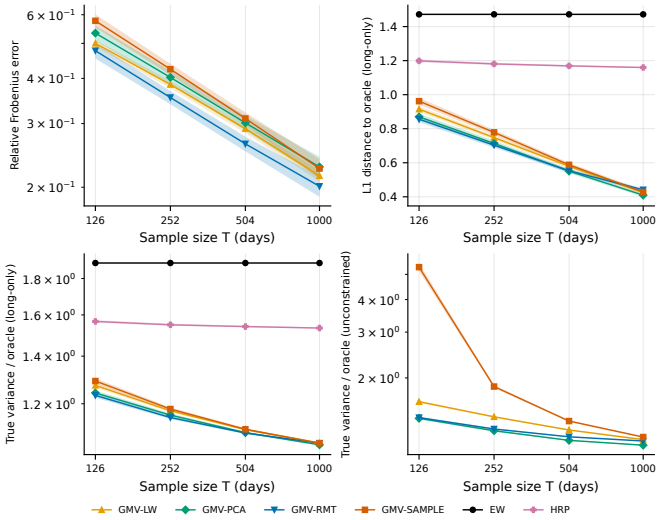


Figure 12: Simulation study with Student-t returns.

Table 7: Metrics on the common period with the S&P 500.

strategy	ann_return	ann_vol	sharpe	sortino	max_drawdown	calmar	cvar_95	annual_turnover
EW	0.144	0.185	0.820	1.147	-0.376	0.382	0.028	
GMV-LW	0.086	0.140	0.658	0.919	-0.335	0.255	0.021	
GMV-PCA	0.081	0.145	0.609	0.850	-0.343	0.236	0.022	
GMV-RMT	0.076	0.147	0.570	0.796	-0.362	0.209	0.022	
GMV-SAMPLE	0.086	0.141	0.658	0.920	-0.333	0.258	0.021	
HRP-RMT	0.119	0.161	0.779	1.081	-0.359	0.332	0.024	
HRP	0.115	0.160	0.762	1.057	-0.356	0.323	0.024	
MV-RMT	0.177	0.218	0.854	1.227	-0.383	0.461	0.033	
MV-SAMPLE	0.175	0.215	0.858	1.225	-0.380	0.460	0.033	
SPX	0.111	0.178	0.679	0.947	-0.339	0.326	0.027	

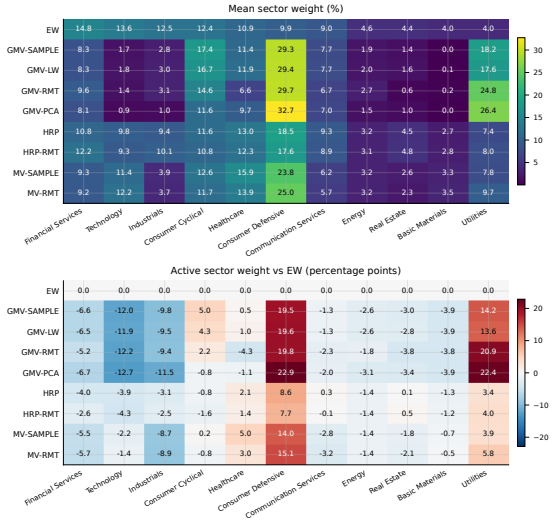


Figure 13: Mean sector exposure of each strategy (current sector labels).

A.3 Extensions (Section 5.7)

Table 8: Extension: EWMA strategies against sample GMV, net of costs.

strategy	ann_vol	ann_return	sharpe	max_drawdown	annual_turnover	mean_pred_vol	realized_over_pred
GMV-SAMPLE	0.130	0.093	0.744	-0.333	2.397	0.110	1.187
GMV-LW	0.130	0.094	0.758	-0.335	2.262	0.109	1.185
GMV-RMT	0.136	0.084	0.663	-0.362	2.195	0.111	1.226
GMV-PCA	0.135	0.092	0.721	-0.343	2.175	0.102	1.319
GMV-EWMA126	0.125	0.101	0.828	-0.295	3.490	0.106	1.183
GMV-EWMA126-RMT	0.130	0.090	0.732	-0.312	3.141	0.107	1.214
GMV-EWMA63	0.123	0.107	0.889	-0.255	5.545	0.100	1.231
GMV-EWMA63-RMT	0.127	0.101	0.825	-0.282	4.938	0.101	1.257

Table 9: Extension: market volatility dynamics, a priori against calibrated.

parameter	a_priori	calibrated
vol_log_sd	0.400	0.437
vol_persistence	0.980	0.981

Table 10: Extension: PCA plus sector-block strategy, net of costs.

strategy	ann_vol	ann_return	sharpe	max_drawdown	annual_turnover	mean_pred_vol	realized_over_pred
GMV-SAMPLE	0.130	0.093	0.744	-0.333	2.397	0.110	1.187
GMV-LW	0.130	0.094	0.758	-0.335	2.262	0.109	1.185
GMV-RMT	0.136	0.084	0.663	-0.362	2.195	0.111	1.226
GMV-PCA	0.135	0.092	0.721	-0.343	2.175	0.102	1.319
GMV-PCA-SECTOR	0.131	0.094	0.753	-0.332	2.233	0.107	1.233

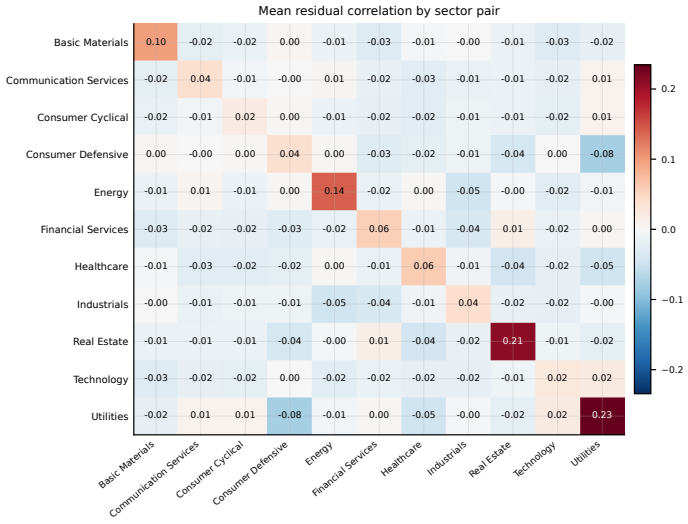


Figure 14: Mean PCA-residual correlation by sector pair.

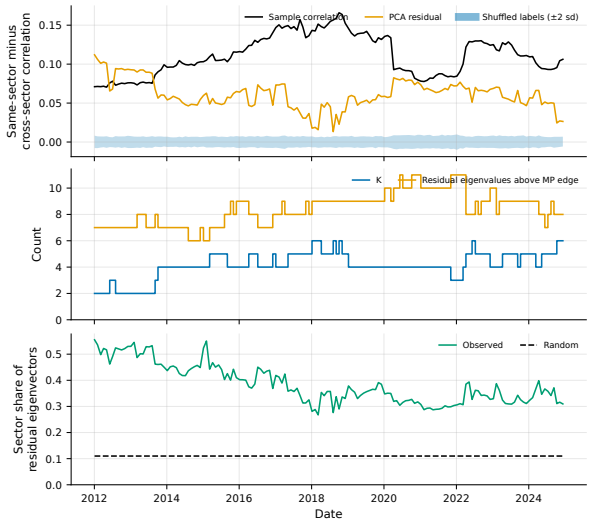


Figure 15: Sector structure of the PCA residual over time.

Table 11: Extension: covariance forecast loss on random portfolios (QLIKE).

family	estimator	n_dates	mean_qlike	mean_log_bias	diff_qlike	se	ci_low	ci_high	p_value	p_holm
equal weight	Ledoit-Wolf	155	0.680	0.485	0.008	0.011	-0.014	0.030	0.465	1.000
equal weight	NLS	155	0.677	0.500	0.005	0.006	-0.006	0.017	0.371	1.000
equal weight	PCA	155	0.668	0.533	-0.004	0.004	-0.013	0.004	0.344	1.000
equal weight	PCA-SECTOR	155	0.666	0.541	-0.006	0.007	-0.019	0.008	0.388	1.000
equal weight	RMT	155	0.670	0.520	-0.002	0.001	-0.004	0.001	0.275	1.000
equal weight	Sample	155	0.672	0.519	0.000					
Dirichlet(1)	Ledoit-Wolf	155	0.662	0.471	0.008	0.011	-0.013	0.028	0.473	1.000
Dirichlet(1)	NLS	155	0.660	0.486	0.005	0.006	-0.006	0.016	0.375	1.000
Dirichlet(1)	PCA	155	0.651	0.518	-0.004	0.004	-0.012	0.004	0.345	1.000
Dirichlet(1)	PCA-SECTOR	155	0.649	0.526	-0.006	0.007	-0.019	0.007	0.389	1.000
Dirichlet(1)	RMT	155	0.653	0.505	-0.001	0.001	-0.004	0.001	0.282	1.000
Dirichlet(1)	Sample	155	0.655	0.505	0.000					
Dirichlet(0.1)	Ledoit-Wolf	155	0.579	0.404	0.005	0.008	-0.010	0.020	0.525	1.000
Dirichlet(0.1)	NLS	155	0.577	0.417	0.003	0.004	-0.004	0.010	0.426	1.000
Dirichlet(0.1)	PCA	155	0.571	0.440	-0.003	0.003	-0.009	0.003	0.344	1.000
Dirichlet(0.1)	PCA-SECTOR	155	0.569	0.446	-0.005	0.005	-0.015	0.006	0.381	1.000
Dirichlet(0.1)	RMT	155	0.572	0.431	-0.001	9.9e-04	-0.003	5.3e-04	0.153	0.766
Dirichlet(0.1)	Sample	155	0.574	0.430	0.000					

Table 12: Extension: quadratic-inverse shrinkage GMV, net of costs.

strategy	ann_vol	ann_return	sharpe	max_drawdown	annual_turnover	mean_pred_vol	realized_over_pred
GMV-SAMPLE	0.130	0.093	0.744	-0.333	2.397	0.110	1.187
GMV-LW	0.130	0.094	0.758	-0.335	2.262	0.109	1.185
GMV-RMT	0.136	0.084	0.663	-0.362	2.195	0.111	1.226
GMV-PCA	0.135	0.092	0.721	-0.343	2.175	0.102	1.319
GMV-NLS	0.130	0.093	0.751	-0.336	2.185	0.110	1.185

Table 13: Extension: inverse-volatility and ERC portfolios, net of costs.

strategy	ann_vol	ann_return	sharpe	max_drawdown	annual_turnover	mean_pred_vol	realized_over_pred
IVOL	0.163	0.153	0.954	-0.372	0.826	0.163	0.998
ERC-SAMPLE	0.157	0.152	0.980	-0.364	0.886	0.156	1.006
ERC-LW	0.157	0.152	0.980	-0.364	0.885	0.154	1.024
ERC-RMT	0.158	0.151	0.970	-0.366	0.876	0.157	1.004
ERC-PCA	0.156	0.151	0.977	-0.364	0.897	0.155	1.007
EW	0.174	0.170	0.993	-0.376	0.869		
GMV-SAMPLE	0.130	0.093	0.744	-0.333	2.397	0.110	1.187

Table 14: Extension: ridge, shrinkage, L1-bounded and elastic-net weights, net of costs.

strategy	ann_vol	ann_return	sharpe	max_drawdown	annual_turnover	mean_leverage	mean_effective_n	mean_pred_vol	realized_over_pred
GMV-RIDGE-0.1	0.129	0.096	0.774	-0.340	2.098	1.000	14.322	0.110	1.172
GMV-RIDGE-1	0.131	0.107	0.840	-0.346	1.524	1.000	30.615	0.117	1.120
GMV-RIDGE-10	0.147	0.138	0.950	-0.359	0.963	1.000	72.500	0.142	1.036
GMV-SHRINK-0.25	0.134	0.113	0.864	-0.344	1.995	1.000	16.800	0.120	1.117
GMV-SHRINK-0.5	0.143	0.132	0.941	-0.355	1.589	1.000	30.833	0.136	1.053
GMV-SHRINK-0.75	0.157	0.152	0.980	-0.365	1.187	1.000	62.898	0.155	1.008
GMV-L1-1.25	0.127	0.082	0.684	-0.336	3.397	1.250	9.864	0.102	1.253
GMV-L1-1.5	0.126	0.081	0.683	-0.331	4.228	1.500	9.021	0.097	1.295
GMV-L1-2	0.125	0.091	0.759	-0.331	6.114	2.000	7.391	0.092	1.367
GMV-ENET-1.5-1	0.127	0.102	0.830	-0.342	2.602	1.479	25.551	0.106	1.201
EW	0.174	0.170	0.993	-0.376	0.869	1.000	100.000		
GMV-SAMPLE	0.130	0.093	0.744	-0.333	2.397	1.000	10.275	0.110	1.187
GMV-LW	0.130	0.094	0.758	-0.335	2.262	1.000	11.888	0.109	1.185
GMV-RMT	0.136	0.084	0.663	-0.362	2.195	1.000	10.008	0.111	1.226

Table 15: Extension: out-of-sample rank IC of the ML signals and the single features.

predictor	mean_ic	std_ic	t_stat	hit_rate	n_dates
RIDGE	0.025	0.226	1.250	0.557	131
RF	0.029	0.192	1.703	0.542	131
mom_12_1	-0.011	0.240	-0.523	0.519	131
ret_1m	-0.026	0.203	-1.493	0.435	131
vol	0.027	0.245	1.279	0.542	131
dollar_volume	0.011	0.138	0.897	0.573	131

Table 16: Extension: ML-fed max-Sharpe portfolios (IC gate failed, shown for completeness) and core references, net of costs.

strategy	ann_vol	ann_return	sharpe	max_drawdown	annual_turnover	mean_effective_n	mean_pred_vol	realized_over_pred
MV-ML-RIDGE-RMT	0.245	0.114	0.562	-0.477	18.590	5.197	0.192	1.277
MV-ML-RF-RMT	0.260	0.097	0.489	-0.489	16.548	4.723	0.207	1.257
EW	0.180	0.149	0.863	-0.376	0.901	100.000		
GMV-SAMPLE	0.137	0.092	0.715	-0.333	2.549	11.203	0.112	1.223
GMV-RMT	0.143	0.084	0.638	-0.362	2.373	10.817	0.113	1.265
MV-SAMPLE	0.210	0.174	0.872	-0.380	5.764	7.085	0.180	1.167
MV-RMT	0.213	0.174	0.858	-0.383	5.970	7.034	0.185	1.148